Table S1: Sites used in this study. Lat= latitude, Long = longitude, NEP = net ecosystem production, GPP = gross primary production, MAT = mean annual temperature, P = total annual precipitation, NA = nutrient availability, Dist = disturbance events used to characterize the stand age of each site where the meaning of the numbers 1-4 are defined in section 1.1 of the supplementaray materials.

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| AU-Cum | -33.613 | 150.722 | 172.67 | 1198.28 | 300.00 | 18.40 | 595.85 | M | 1 | 2012-2014 | - |
| AU-How | -12.494 | 131.152 | 435.47 | 2070.71 | 110.00 | 26.99 | 1681.15 | M | 1 | 2001-2014 | - |
| AU-Rob | -17.117 | 145.630 | 657.52 | 2116.39 | 198.00 | 21.72 | 2193.83 | M | 1 | 2014 | - |
| AU-Tum | -35.656 | 148.151 | 633.15 | 3328.78 | 83.00 | 9.78 | 889.26 | M | 1 | 2001-2006 | [21] |
| AU-Wom | -37.422 | 144.094 | 681.83 | 1737.04 | 32.00 | 11.97 | 786.50 | M | 1 | 2010-2014 | - |
| BE-Bra | 51.309 | 4.520 | 118.73 | 1426.98 | 78.00 | 10.82 | 824.67 | L | 1 | 1996-2013 | [15] |
| BE-Vie | 50.305 | 5.998 | 454.14 | 1755.48 | 94.00 | 8.34 | 966.29 | M | 1 | 1996-2014 | [3] |
| BR-Sa1 | -2.856 | -54.958 | -26.53 | 3342.92 | 300.00 | 25.96 | 2596.91 | L | 1 | 2002-2011 | - |
| BR-Sa3 | -3.018 | -54.971 | 105.85 | 3045.54 | 2 | 300.00 | 1408.04 | L | 1 | 2000-2004 | - |
| CA-Ca1 | 49.867 | -125.333 | 391.49 | 2232.29 | 60.00 | 8.39 | 1298.95 | L | 1 | 1997-2005 | [52] |
| CA-Ca2 | 49.870 | -125.291 | -518.73 | 588.30 | 2.00 | 8.89 | 1214.01 | L | 1 | 2000-2005 | [34] |
| CA-Ca3 | 49.534 | -124.900 | 11.55 | 1342.57 | 16.00 | 9.66 | 1543.54 | L | 1 | 2001-2005 | [37] |
| CA-Gro | 48.216 | -82.155 | 117.79 | 1065.12 | 78.00 | 3.60 | 770.32 | M | 1 | 2003-2014 | [58] |
| CA-Man | 55.879 | -98.480 | 22.56 | 687.54 | 161.00 | -1.11 | 310.38 | L | 1 | 1994-2003 | [39] |
| CA-NS1 | 55.879 | -98.483 | 89.03 | 739.28 | 154.00 | 0.22 | 215.28 | L | 1 | 2002-2005 | [26] |

*Continued on next page*

Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CA-NS2 | 55.905 | -98.524 | 172.45 | 948.37 | 73.00 | -3.70 | 275.29 | L | 1 | 2001-2005 | [26] |
| CA-NS3 | 55.911 | -98.382 | 60.76 | 570.64 | 39.00 | -2.93 | 196.49 | L | 1 | 2001-2005 | [26] |
| CA-NS5 | 55.863 | -98.485 | 80.68 | 695.77 | 23.00 | -3.42 | 220.66 | L | 1 | 2001-2005 | [26] |
| CA-NS6 | 55.916 | -98.964 | 15.21 | 421.48 | 14.00 | -1.11 | 201.88 | L | 1 | 2001-2005 | [26] |
| CA-NS7 | 56.635 | -99.948 | -103.98 | 422.34 | 6.00 | -2.09 | 235.75 | L | 1 | 2002-2005 | [26] |
| CA-Oas | 53.628 | -106.197 | 122.27 | 1113.43 | 80.00 | 1.96 | 479.07 | L | 1 | 1997-2005 | [9] |
| CA-Obs | 53.987 | -105.117 | 38.26 | 841.18 | 112.00 | 1.12 | 495.74 | L | 1 | 1999-2005 | [36] |
| CA-Ojp | 53.916 | -104.692 | 46.83 | 572.34 | 88.00 | 1.40 | 454.21 | L | 1 | 1999-2005 | [4] |
| CA-Qcu | 49.267 | -74.036 | -146.72 | 321.05 | 4.00 | 1.16 | 930.16 | L | 1 | 2001-2006 | [22] |
| CA-Qfo | 49.692 | -74.342 | 3.47 | 648.33 | 102.00 | 1.13 | 939.19 | L | 1 | 2003-2010 | [6] |
| CA-SJ1 | 53.908 | -104.655 | -25.48 | 373.52 | 10.00 | 0.57 | 734.31 | L | 1 | 2001-2005 | [33] |
| CA-SJ2 | 53.944 | -104.649 | -130.15 | 89.81 | 2.00 | -0.03 | 674.45 | L | 1 | 2003-2005 | [14] |
| CA-SJ3 | 53.875 | -104.649 | 90.34 | 535.52 | 30.00 | 1.80 | 1137.16 | L | 1 | 2004-2005 | [27] |
| CA-TP1 | 42.660 | -80.559 | 183.45 | 1301.86 | 9.00 | 9.03 | 1090.20 | M | 1 | 2002-2014 | [2] |
| CA-TP3 | 42.706 | -80.348 | 451.11 | 1682.59 | 37.00 | 8.91 | 1145.72 | M | 1 | 2002-2014 | [2] |
| CA-TP4 | 42.710 | -80.357 | 90.00 | 1508.94 | 70.00 | 8.52 | 980.58 | M | 1 | 2002-2014 | [2] |
| CA-TPD | 42.635 | -80.557 | 299.88 | 1499.02 | 98.00 | 9.66 | 920.81 | M | 1 | 2012-2014 | [2] |

*Continued on next page*

Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CA-WP1 | 54.953 | -112.467 | 250.03 | 693.32 | 136.00 | 1.96 | 439.89 | L | 1 | 2003-2005 | - |
| CH-Dav | 46.815 | 9.855 | 116.39 | 1238.17 | 222.00 | 3.54 | 849.99 | L | 1 | 1997-2015 | [81] |
| CH-Lae | 47.478 | 8.365 | 662.11 | 1931.91 | 184.00 | 7.72 | 1195.59 | M | 1 | 2004-2015 | [20] |
| CN-Cha | 42.402 | 128.095 | 268.85 | 1496.07 | 300.00 | 4.35 | 466.14 | H | 1 | 2003-2005 | [30] |
| CN-Din | 23.173 | 112.536 | 469.94 | 1462.56 | 96.00 | 20.40 | 1411.43 | L | 1 | 2003-2005 | [80] |
| CN-Qia | 26.741 | 115.058 | 573.53 | 1812.07 | 19.00 | 18.24 | 1169.89 | L | 1 | 2003-2005 | - |
| CZ-BK1 | 49.502 | 18.536 | 831.75 | 1844.97 | 31.00 | 6.53 | 1102.62 | M | 1 | 2000-2012 | [61] |
| DE-Bay | 50.141 | 11.866 | -45.49 | 1354.17 | 54.00 | 6.30 | 1241.17 | L | 1 | 1996-1999 | [75] |
| DE-Hai | 51.079 | 10.453 | 564.43 | 1662.84 | 254.00 | 8.47 | 756.57 | H | 1 | 2000-2012 | [40] |
| DE-Har | 47.934 | 7.6009 | 618.93 | 1530.12 | 42.00 | 11.12 | 619.00 | M | 1 | 2005-2006 | [8] |
| DE-Lkb | 49.099 | 13.304 | -267.09 | 510.12 | 2.00 | 5.48 | 873.63 | M | 4 | 2009-2013 | [44] |
| DE-Lnf | 51.328 | 10.367 | 594.26 | 1685.98 | 117.00 | 8.07 | 541.38 | M | 1 | 2002-2012 | - |
| DE-Meh | 51.275 | 10.655 | -3.25 | 1136.89 | 2.00 | 8.80 | 520.43 | H | 1 | 2003-2006 | - |
| DE-Obe | 50.783 | 13.719 | 372.53 | 1805.06 | 76.00 | 6.49 | 1044.06 | M | 1 | 2008-2014 | - |
| DE-Tha | 50.963 | 13.566 | 613.99 | 1980.87 | 118.00 | 8.91 | 841.43 | M | 1 | 1996-2014 | [7] |
| DE-Wet | 50.453 | 11.457 | 82.15 | 1529.44 | 54.00 | 6.41 | 1021.64 | L | 1 | 2002-2006 | [1] |
| DK-Sor | 55.485 | 11.644 | 167.33 | 1952.19 | 85.00 | 8.40 | 854.38 | H | 1 | 1996-2012 | [59] |

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Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ES-ES1 | 39.346 | -0.318 | 404.01 | 1432.53 | 116.00 | 17.41 | 607.96 | L | 1 | 1999-2006 | [66] |
| ES-LMa | 39.941 | -5.773 | 122.29 | 1069.11 | 148.00 | 16.16 | 778.24 | L | 1 | 2004-2006 | - |
| FI-Hyy | 61.847 | 24.29 | 219.40 | 1118.01 | 46.00 | 4.40 | 607.38 | L | 1 | 1996-2014 | [68] |
| FI-Sod | 67.361 | 26.637 | -101.60 | 598.89 | 83.00 | 0.77 | 534.58 | L | 1 | 2000-2006 | [71] |
| FR-Fon | 48.476 | 2.780 | 651.50 | 1835.12 | 161.00 | 11.25 | 669.27 | H | 1 | 2005-2014 | [49] |
| FR-Hes | 48.674 | 7.0646 | 333.77 | 1599.72 | 37.00 | 10.12 | 957.32 | L | 1 | 1997-2006 | [28] |
| FR-LBr | 44.720 | -0.770 | 299.94 | 1726.74 | 34.00 | 13.40 | 939.40 | L | 1 | 1996-2006 | [5] |
| FR-Pue | 43.741 | 3.595 | 233.42 | 1214.59 | 64.00 | 13.74 | 937.66 | M | 1 | 2000-2013 | [60] |
| GF-Guy | 5.278 | -52.924 | 154.65 | 3724.44 | 300.00 | 25.61 | 3110.49 | L | 1 | 2004-2012 | - |
| IL-Yat | 31.344 | 35.051 | 201.20 | 649.62 | 39.00 | 18.01 | 265.50 | M | 1 | 2001-2006 | [29] |
| IT-Col | 41.849 | 13.588 | 537.18 | 1300.52 | 180.00 | 7.28 | 999.41 | H | 1 | 1996-2006 | [74] |
| IT-Cp2 | 41.704 | 12.357 | 590.85 | 2031.64 | 63.00 | 15.82 | 915.08 | M | 1 | 2012-2014 | - |
| IT-Cpz | 41.705 | 12.376 | 488.78 | 1936.53 | 56.00 | 16.19 | 742.46 | L | 1 | 1997-2006 | [72] |
| IT-La2 | 45.9542 | 11.2853 | 1049.37 | 2108.42 | 89.00 | 6.76 | 1177.20 | H | 1 | 2000-2002 | - |
| IT-LMa | 45.581 | 7.154 | 595.15 | 909.33 | 71.00 | 14.10 | 804.15 | H | 1 | 2003-2006 | [47] |
| IT-Noe | 40.606 | 8.151 | 205.46 | 1122.14 | 49.00 | 16.31 | 586.35 | M | 1 | 2004-2012 | [62] |
| IT-Non | 44.689 | 11.088 | 481.36 | 1346.20 | 10.00 | 13.59 | 1221.79 | H | 1 | 2001-2006 | [54] |

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Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| IT-PT1 | 45.200 | 9.061 | 494.79 | 1477.01 | 13.00 | 14.27 | 540.29 | L | 1 | 2002-2004 | [50] |
| IT-Ren | 46.587 | 11.434 | 693.16 | 1498.57 | 188.00 | 4.57 | 904.97 | M | 1 | 1998-2013 | - |
| IT-Ro1 | 42.408 | 11.930 | 229.25 | 1550.23 | 10.00 | 15.47 | 851.96 | H | 4 | 2000-2008 | [63] |
| IT-Ro2 | 42.390 | 11.920 | 686.27 | 1443.09 | 19.00 | 15.36 | 802.84 | H | 4 | 2002-2012 | [70] |
| IT-SR2 | 43.732 | 10.291 | 351.58 | 2343.56 | 64.00 | 15.47 | 1328.66 | L | 1 | 2013-2014 | - |
| IT-SRo | 43.727 | 10.284 | 369.82 | 1931.90 | 54.00 | 15.38 | 820.96 | L | 1 | 1999-2012 | [11] |
| JP-Tak | 36.146 | 137.423 | 308.20 | 1087.49 | 72.00 | 6.60 | 2.26 | H | 1 | 1999-2004 | [79] |
| JP-Tef | 45.056 | 142.106 | -307.58 | 979.21 | 84.00 | 5.88 | 965.85 | L | 4 | 2001-2011 | [69] |
| JP-Tom | 42.739 | 141.514 | 331.19 | 1710.27 | 48.00 | 6.28 | 1065.99 | L | 1 | 2001-2003 | [31] |
| MY-PSO | 2.973 | 102.306 | 129.89 | 2554.45 | 106.00 | 25.34 | 1864.85 | L | 1 | 2003-2009 | [35] |
| NL-Loo | 52.166 | 5.743 | 450.75 | 1637.40 | 106.00 | 10.08 | 803.22 | M | 1 | 1996-2014 | [18] |
| PA-SPn | 9.318 | -79.634 | 441.80 | 2088.30 | 7.00 | 25.16 | 2073.59 | M | 1 | 2007-2009 | - |
| PT-Esp | 38.639 | -8.601 | 629.21 | 1842.15 | 12.00 | 16.36 | 592.53 | L | 1 | 2002-2006 | - |
| PT-Mi1 | 38.540 | -8.000 | -39.88 | 828.81 | 88.00 | 15.80 | 243.84 | L | 1 | 2003-2005 | - |
| RU-Fyo | 56.461 | 32.922 | -137.79 | 1496.93 | 236.00 | 5.23 | 560.63 | L | 1 | 1998-2013 | [41] |
| SE-Fla | 64.112 | 19.456 | 59.17 | 704.12 | 37.00 | 3.06 | 703.79 | L | 1 | 1996-2002 | [75] |
| SE-Nor | 60.086 | 17.479 | -75.49 | 1155.43 | 105.00 | 6.06 | 703.73 | L | 1 | 1996-2005 | [42] |

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Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SE-Sk1 | 60.125 | 17.918 | -186.43 | 461.16 | 2.00 | 9.30 | 0.59 | H | 1 | 2005 | [23] |
| SE-Sk2 | 60.129 | 17.840 | 66.34 | 674.11 | 33.00 | 6.69 | 0.58 | H | 1 | 2004-2005 | - |
| UK-Gri | 56.607 | -3.798 | 515.53 | 1710.60 | 21.00 | 7.48 | 1115.84 | H | 1 | 1997-2006 | [48] |
| UK-Ham | 51.120 | -0.860 | 595.09 | 2111.61 | 64.00 | 10.47 | 796.70 | H | 1 | 2004-2005 | [78] |
| US-Bar | 44.064 | -71.288 | 367.15 | 1102.15 | 128.00 | 7.15 | 1254.52 | M | 1 | 2004-2005 | [38] |
| US-Blo | 38.895 | -120.632 | 176.75 | 1965.20 | 13.00 | 10.97 | 1526.95 | L | 1 | 1997-2007 | [24] |
| US-Bn1 | 63.919 | -145.378 | 268.68 | 524.40 | 83.00 | -1.10 | 249.15 | L | 1 | 2003 | [46] |
| US-Bn2 | 63.919 | -145.378 | 239.81 | 512.99 | 16.00 | -0.48 | 249.15 | L | 1 | 2003 | [46] |
| US-Bn3 | 63.922 | -145.744 | 91.85 | 238.46 | 4.00 | -0.71 | 249.15 | L | 1 | 2003 | [46] |
| US-Dk3 | 35.978 | -79.094 | 756.04 | 2245.52 | 21.00 | 14.95 | 1037.37 | L | 1 | 2001-2005 | [67] |
| US-GBT | 41.365 | -106.239 | 528.98 | 718.88 | 176.00 | -0.03 | 424.41 | M | 1 | 1999-2006 | - |
| US-GLE | 41.366 | -106.239 | -83.52 | 618.81 | 184.00 | -0.03 | 1405.35 | M | 1 | 2004-2014 | - |
| US-Ha1 | 42.537 | -72.171 | 199.79 | 1506.46 | 96.00 | 8.32 | 1139.66 | L | 1 | 1991-2012 | [73] |
| US-Ho1 | 45.204 | -68.740 | 255.50 | 1468.32 | 206.00 | 6.60 | 815.29 | L | 1 | 1996-2014 | [32] |
| US-Ho2 | 45.209 | -68.747 | 231.09 | 1342.73 | 208.00 | 6.51 | 787.37 | L | 1 | 1999-2009 | [32] |
| US-Los | 46.082 | -89.979 | 48.07 | 783.45 | 35.00 | 4.92 | 729.34 | L | 1 | 2000-2014 | - |
| US-LPH | 42.541 | -72.184 | 368.35 | 1233.06 | 98.00 | 7.33 | 2520.50 | M | 1 | 2002-2005 | [10] |

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Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| US-Me2 | 44.452 | -121.557 | 604.98 | 1616.96 | 94.00 | 7.31 | 457.49 | M | 1 | 2002-2014 | [43] |
| US-Me3 | 44.315 | -121.607 | 48.02 | 781.76 | 20.00 | 8.20 | 367.11 | M | 1 | 2004-2005 | [77] |
| US-Me5 | 44.437 | -121.566 | 137.85 | 818.46 | 24.00 | 7.89 | 412.90 | M | 1 | 2000-2002 | [43] |
| US-Me6 | 44.323 | -121.607 | 180.66 | 882.75 | 22.00 | 7.71 | 409.02 | M | 1 | 2010-2012 | [65] |
| US-MMS | 39.323 | -86.413 | 423.08 | 1718.92 | 95.00 | 12.41 | 1084.11 | M | 1 | 1999-2014 | [64] |
| US-MOz | 38.744 | -92.199 | 353.74 | 1424.79 | 78.00 | 13.52 | 873.63 | L | 1 | 2004-2014 | - |
| US-NC1 | 35.811 | -76.711 | -178.28 | 1629.06 | 2.00 | 15.59 | 1056.09 | M | 1 | 2005-2009 | [57] |
| US-NC2 | 35.803 | -76.667 | 936.24 | 2724.41 | 14.00 | 15.92 | 1272.59 | M | 1 | 2005-2009 | [55] |
| US-NR1 | 40.032 | -105.546 | 168.94 | 846.05 | 110.00 | 2.28 | 726.36 | L | 1 | 1998-2014 | [51] |
| US-Oho | 41.554 | -83.843 | 827.98 | 1776.81 | 50.00 | 10.52 | 818.23 | L | 1 | 2004-2013 | [56] |
| US-PFa | 45.945 | -90.272 | 9.37 | 959.85 | 150.00 | 5.72 | 600.94 | H | 1 | 1995-2014 | [16] |
| US-Prr | 65.123 | -147.487 | 123.43 | 496.91 | 98.00 | -2.02 | 382.16 | L | 1 | 2010-2014 | [53] |
| US-SO2 | 33.373 | -116.622 | 59.07 | 283.07 | 2.00 | 14.43 | 532.54 | M | 4 | 1997-2006 | [45] |
| US-SO3 | 33.377 | -116.622 | 86.86 | 341.03 | 2.00 | 14.79 | 454.00 | M | 4 | 1997-2006 | [45] |
| US-SP1 | 29.738 | -82.218 | 161.84 | 1722.32 | 63.00 | 19.73 | 53.86 | L | 1 | 2000-2005 | [12] |
| US-SP2 | 29.764 | -82.244 | 162.00 | 2178.24 | 4.00 | 19.70 | 1118.41 | L | 1 | 1998-2004 | [12] |
| US-SP3 | 29.754 | -82.163 | 542.14 | 1875.13 | 12.00 | 20.14 | 1075.60 | L | 1 | 1999-2004 | [12] |

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Table S1 – *Continued from previous page*

| Site | Lat [°N] | Long [°E] | NEP [gC m$^{-2}$ y$^{-1}$] | GPP [gC m$^{-2}$ y$^{-1}$] | Age [years] | MAT [°C] | P [mm y$^{-1}$] | NA | Dist | Years | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|
| US-SRM | 31.821 | -110.866 | -24.81 | 283.18 | 201.00 | 19.01 | 333.19 | L | 1 | 2004-2014 | - |
| US-Syv | 46.242 | -89.347 | 80.38 | 1171.41 | 300.00 | 3.92 | 692.48 | L | 1 | 2001-2014 | [17] |
| US-UMB | 45.559 | -84.713 | 268.69 | 1299.05 | 93.00 | 7.15 | 613.06 | L | 1 | 2000-2014 | [25] |
| US-UMd | 45.562 | -84.697 | 316.98 | 1332.29 | 90.00 | 7.36 | 727.64 | L | 1 | 2007-2014 | - |
| US-WBW | 35.958 | -84.287 | 815.94 | 1453.80 | 110.00 | 14.64 | 1.52 | L | 1 | 1995-1999 | [76] |
| US-WCr | 45.805 | -90.079 | 282.37 | 1173.52 | 96.00 | 5.77 | 695.24 | M | 1 | 1999-2014 | [13] |
| US-Wrc | 45.820 | -121.952 | 347.08 | 1625.95 | 300.00 | 9.41 | 2139.87 | M | 1 | 1998-2006 | - |
| VU-Coc | -15.443 | 167.192 | 358.24 | 3499.22 | 20.00 | 24.71 | 2734.43 | H | 1 | 2001-2004 | - |
| ZM-Mon | -15.434 | 23.253 | -147.63 | 1566.22 | 88.00 | 22.20 | 682.80 | M | 4 | 2000-2009 | - |

## 1. Supplementaray methods

### *1.1. Characterization of forest age estimates*

In order to define forest age, we looked at four different scenarii that were reported in the Biological, Ancillary, Disturbance and Metadata database (BADM), in publications, or provided by the Principal Investigators:

1. Complete stand replacement events: reports of a clear cut, a crop to tree cover (afforestation) transition, or a stand replacing fire event (either natural or a controlled experiment). In these cases the event is reported as to have had covered the whole area under the observational footprint of the eddy covariance tower. We used this information to set the age of the site.
2. Minor stand disturbance events: for some sites, the BADM reports insect outbreaks or wind throws covering minor parts of the stand and footprint of the eddy covariance towers. This information is rarely qualitative, but most of the time refers to low/minor intensity disturbance. In these cases, we did not adjust the age of the stand.
3. Marginal logging events: the BADM reports around 25 sites in which selective logging event harvesting occurred. In four cases, there was some quantitative information on the amount of biomass removed(e.g. BR-Sa3: selectively logged in September 2001, where only 5% of the aboveground biomass were removed; US-SP1: 27% of basal area removed). In fact, most of the time, only qualitative information on the intensity of these disturbance (i.e. minor or low) is provided or the PIs only reported that a disturbance event occurred but with no information on its intensity. Therefore, it was not possible to attribute an age change as there was no information on tree density, or age of the trees logged. Like for the minor stand disturbance events, we opted for not adjusting the age of the stand. Other qualitative disturbance intensity reports fall under case #4
4. Partial substantial disturbance: there are cases where the sites (i.e. 5 sites) undergo intense disturbance events (i.e. fire) impacting a majority of the trees, which are reported in the BADM, or publications. In all these cases, we consider that the age is the difference between the flux observation year and the year of disturbance. When the disturbance occurred prior to the observational period there is no other record to resource than the site reports of a large and intense disturbance. Perhaps these

are the cases where the highest uncertainties in age emerge that could contribute to further improvements in the models.

### 1.2. Eddy-covariance processing

We used the night time partitioning, variable ustar, and reference flux observation versions. Data processing included: storage-correction; spike and u*-filtering; flux partitioning [61]. Only sites with more than 80% of the original or good quality gap-filled data were included in this analysis.

### 1.3. Latent heat flux filtering and GPP′ computation

We filtered out daily LE estimates below the 10% quantile within site and where air temperature was lower than 10°C and computed GPP′ as the annual sum of the ratio between latent heat flux (LE) and the square root of vapor pressure deficit (VPD) computed at daily day-time scale.

### 1.4. Spuriousness analysis

We computed the coefficient of spurious correlation (Rsc) adapted from :

- GPP = NEE + ER
- NEE= -GPP + ER
- ER = NEE + GPP

The spurious correlation coefficient can then be calculated as follows:

- Rsc= -var(ER)/ [(var(GPP) + var(ER))$^{0.5}$ x (var(NEE) + var(ER))$^{0.5}$]

We found that Rsc is equal to -0.58 to -0.57 for site-years and site-average, respectively. The Rsc estimates suggest that there is a moderate spurious correlation between NEE and GPP at annual scale, although we show that Rsc varies from low to moderate amongst sites with a mean value of -0.48. This is suggesting that the correlation between GPP and NEP at annual scales was unlikely to be an artifact from the partitioning.

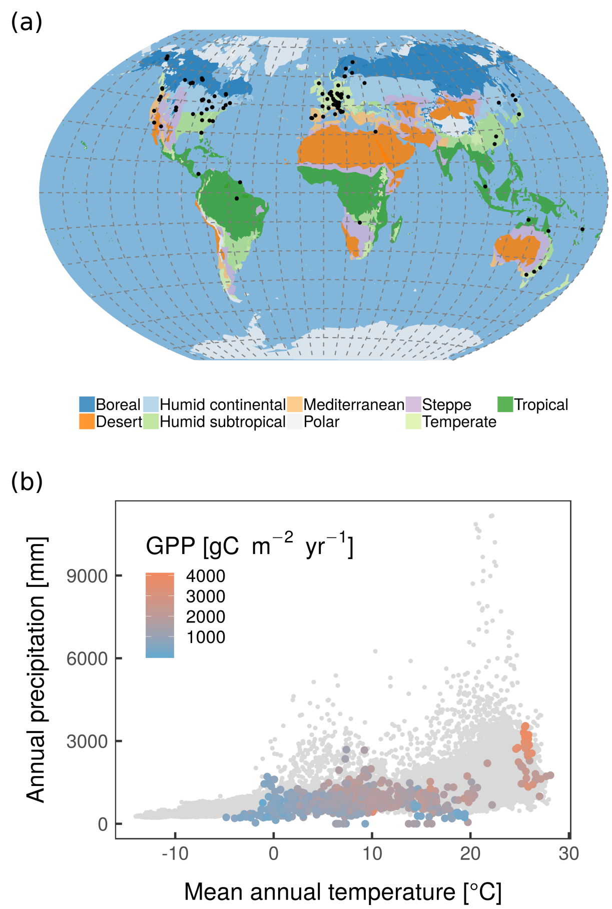


Fig. S1: Geographical (Fig. S1a) and climatological (Fig. S1b) distribution of the sites used in the analysis. In S1a the black dots indicate the location of the different FLUXNET sites. In S1b the color code in the legend represents annual GPP [gC m$^{-2}$ y$^{-1}$]. The data points colored according to the color bar represent the sites precipitation-temperature combinations (climate-space) of the measurements at annual scale while the gray dots represent the climate space of the global forest ecosystems. The climate classification system is derived from Köppen Geiger A2 Scenario 2001-2025.

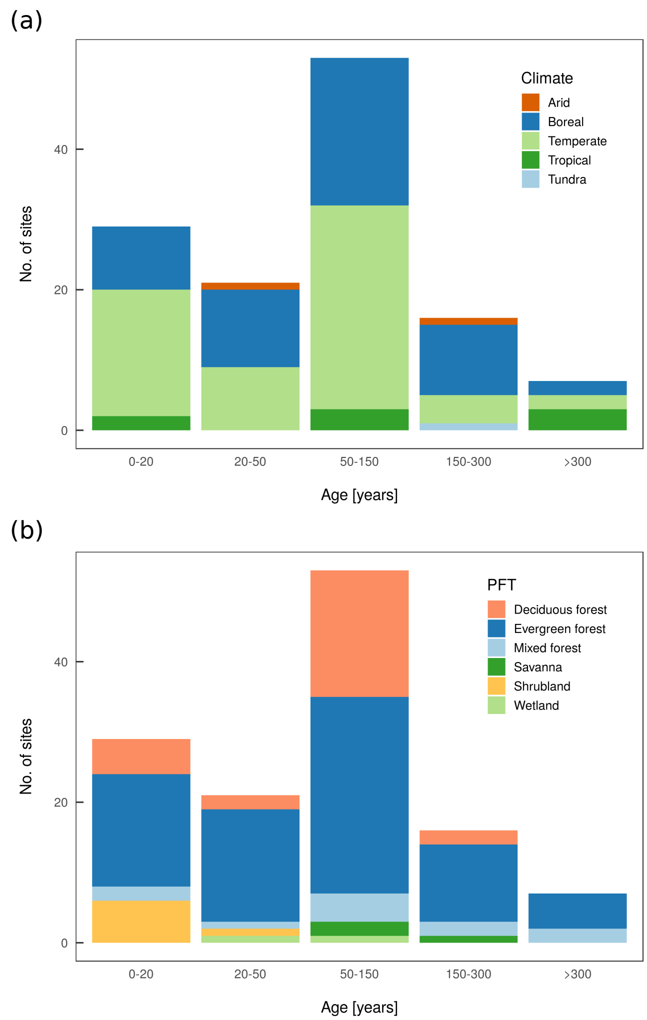


Fig. S2: Histogram of forest age from young to old-growth forests in our dataset by (Fig. S2a) major climate zones and (Fig. S2b) plant functional types.

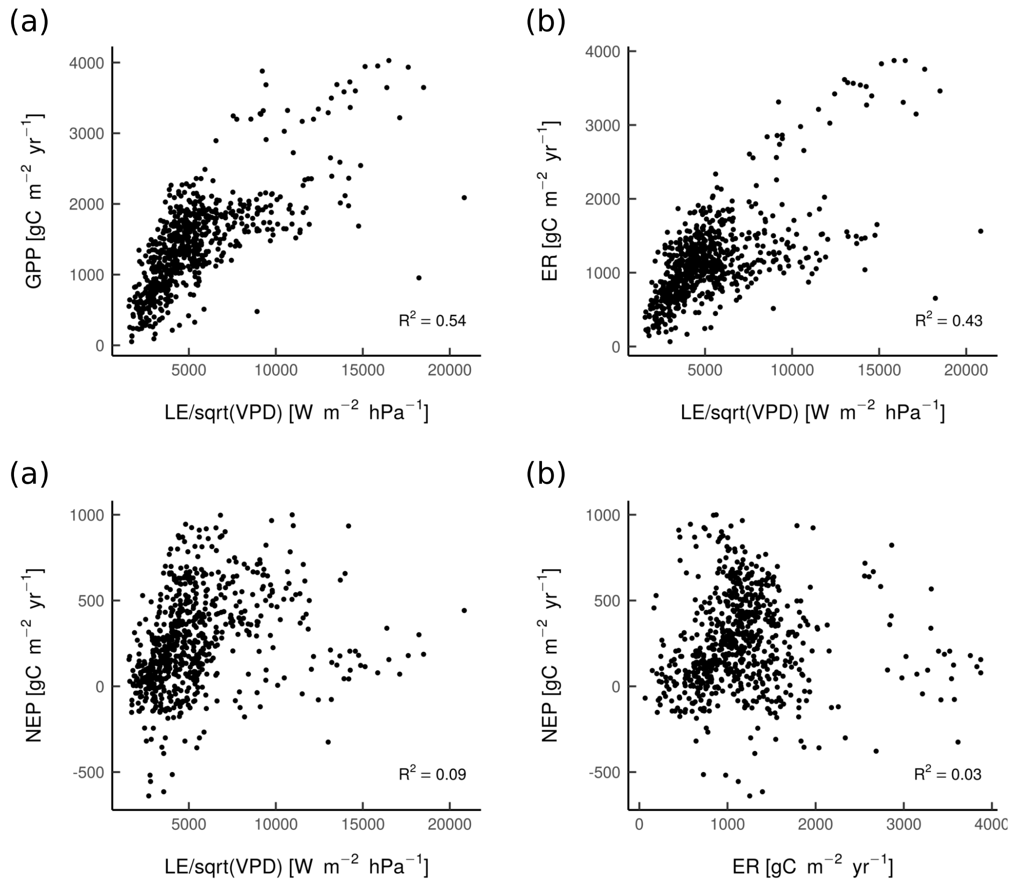


Fig. S3: Correlation between LE/sqrt(VPD) or GPP′ and GPP (Fig. S3a), correlation between GPP′ and ecosystem respiration (ER) (Fig. S3b), correlation GPP′ and NEP (Fig. S3c), and correlation between ER and NEP (Fig. S3d).

Table S2: Model performance comparision between Amiro, Coursolle and Tang models. The Nash-Sutcliffe model efficiency coefficient is reported in the table.

| | Site-years | Site-average |
|---|---|---|
| Amiro | 0.62 | 0.71 |
| Coursolle | 0.38 | 0.41 |
| Tang | 0.40 | 0.44 |

Table S3: Results of the variable selection procedure. Only the five best variables were kept in the final Random Forest model. The Z-score is computed by dividing the average accuracy loss when permuting variables by its standard deviation.

| | mean Z-score | median Z-score | min Z-score | max Z-score |
|---:|---:|---:|---:|---:|
| f(Age) | 33.06 | 33.40 | 30.39 | 34.80 |
| Forest age | 24.43 | 24.43 | 22.46 | 27.14 |
| $\mathrm{GPP}'$ | 23.23 | 23.31 | 20.47 | 24.93 |
| $\mathrm{N}_{deposition}$ | 21.35 | 21.33 | 19.99 | 22.66 |
| MAT | 20.47 | 20.60 | 18.19 | 21.92 |
| Silt content | 18.99 | 18.98 | 17.34 | 20.61 |
| MAT trend | 18.02 | 18.13 | 14.37 | 20.20 |
| P trend | 17.18 | 17.22 | 14.18 | 18.72 |
| Sand content | 16.05 | 16.16 | 14.04 | 17.40 |
| Clay content | 15.82 | 15.94 | 14.31 | 17.45 |
| Global radiation | 15.39 | 15.21 | 14.23 | 18.48 |
| P | 12.40 12.27 | 10.62 | 14.32 | |
| Nutrient availability | 11.80 | 11.70 | 10.52 | 13.96 |
| Forest management | 8.19 | 8.20 | 7.11 | 9.00 |
| P anomaly | 3.68 | 3.67 | 1.00 | 5.97 |
| MAT anomaly | 1.40 | 1.38 | -0.80 | 3.15 |

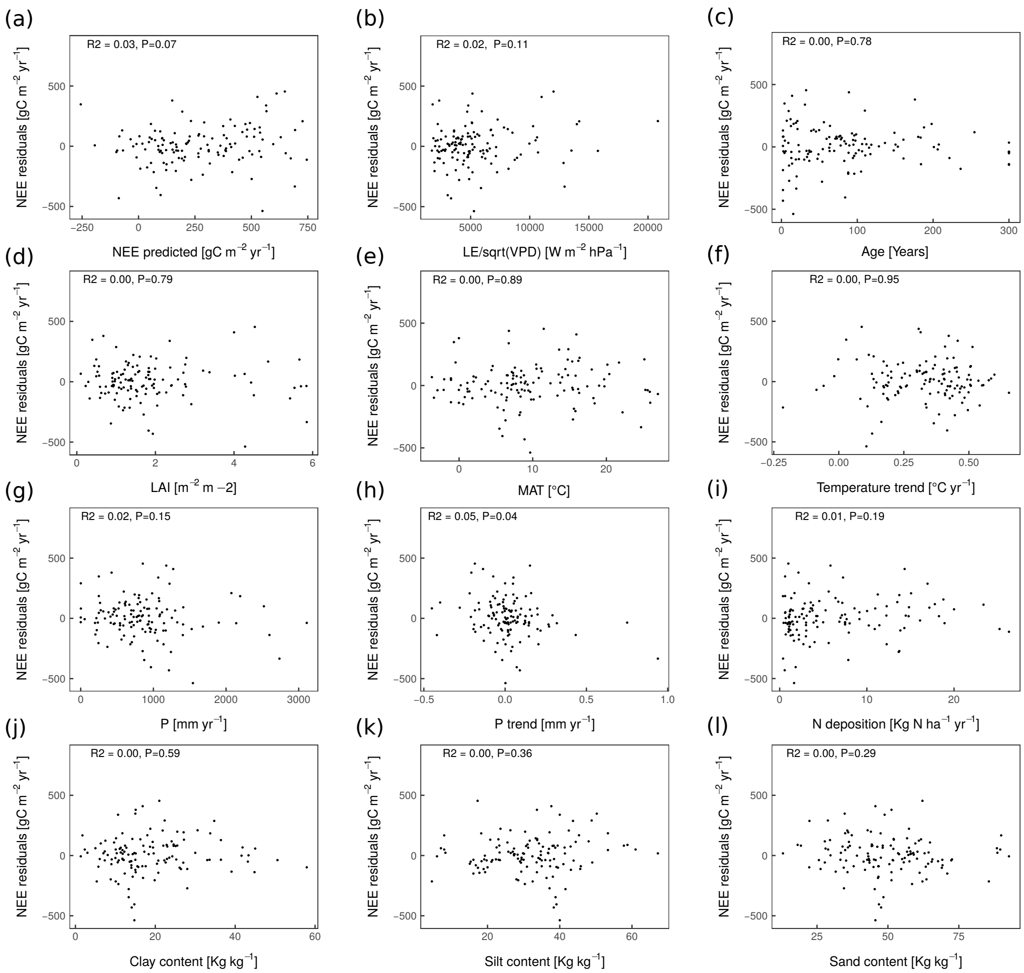


Fig. S4: NEP Residuals against (a) the modeled estimates and variables considered in the analysis: (b) GPP′, (c) age, (d) LAI, (e) MAT, (f) temperature trend, (g) P, (h) precipitation trend, (i) N deposition, (j) clay content, (k) silt content, and (l) sand content. The dots are sites (n=123).

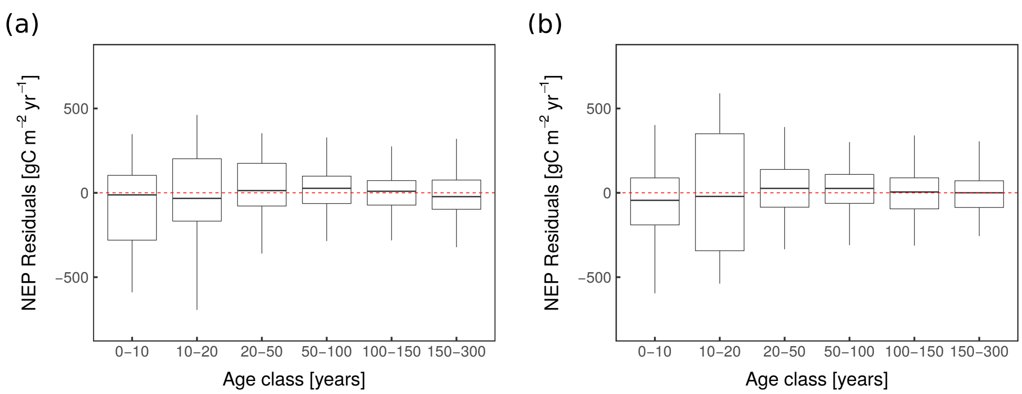


Fig. S5: Model residuals per age class of (Fig. S5a) a RF model including GPP′ as a predictive variable, i.e. NEP = f(age, f(Age), GPP′, MAT, $N_{deposition}$) and (Fig. S5b) a RF model excluding GPP′ and MAT as a predictive variable, i.e. NEP = f(age, f(Age), $N_{deposition}$).

Table S4: Changes in model performance by subtracting forest age (not using f(Age)) from model formulation. NEP = f(age, GPP′, MAT, $N_{deposition}$). $R^2$ = coefficient of determination; MEF = modelling efficiency; RMSE = root mean squared error; MAE = mean absolute error; total n = 699 for site-years and n= 123 for site-average.

| | **Δ $R^2$** | **Δ MEF** | **Δ RMSE (gC $m^{-2}$ $y^{-1}$)** | **Δ MAE (gC $m^{-2}$ $y^{-1}$)** |
|---|---|---|---|---|
| **Site-years** | | | | |
| Full model | 0.29 | 0.29 | 248.23 | 189.05 |
| (-) Age | -0.08 | -0.10 | +16.01 | +11.75 |
| **Site-average** | | | | |
| Full model | 0.31 | 0.31 | 242.37 | 180.85 |
| (-) Age | -0.10 | -0.11 | +18.11 | +22.71 |

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